

Differentiating a local mollification through its kernel

MathlibAnnex.WeakGradient.localMollification_fderiv_apply

theorem

Writes a directional derivative without differentiating the locally integrable input.

Statement

Let E be a finite-dimensional real normed vector space with its Borel measurable structure and an additive Haar measure μ . Fix $R > 0$. For $k \in \mathbb{N}$, let $\varphi_k : E \rightarrow \mathbb{R}$ be the normalized smooth bump centered at zero with inner and outer radii

$$a_k = \frac{R}{2(k+2)}, \quad r_k = \frac{R}{k+2}.$$

It is nonnegative, has integral one, and has compact support in $\overline{B}(0, r_k)$. If $v : E \rightarrow \mathbb{R}$ is locally integrable, put

$$v_k(y) := (\varphi_k * v)(y) = \int_E \varphi_k(t) v(y - t) d\mu(t).$$

Then, for every $k, y \in E$, and direction $e \in E$,

$$Dv_k(y)e = \int_E v(z) D\varphi_k(y - z)e d\mu(z).$$

Assumptions

Only local integrability of v is assumed. The theorem does not require a weak equation, global integrability, Lipschitz continuity, or differentiability of v . The dimension may be zero. The kernel is the specified normalized bump for these radii; no extra scaling formula for a universal kernel is assumed.

Conclusion

The derivative is a genuine Fréchet derivative of the mollification, and the displayed scalar integral is integrable. Differentiation falls on the smooth compactly supported kernel, not on v .

In the local application, $v = \mathbf{1}_K u$ with K compact and contained in an open set on which u is locally integrable. Local integrability gives $\int_K |u| d\mu < \infty$, and measurability of compact K gives integrability of $\mathbf{1}_K u$ on E , hence the hypothesis of this theorem. Separately, the rank-one trace identity $\mathrm{tr}(z \mapsto L(z)e) = L(e)$ explains why the same directional kernel derivative occurs in the divergence test used later; it is not an assumption on v .

Proof route

Apply the compact-kernel differentiation theorem for convolution, evaluate its operator-valued integral in direction e , and change variables by the Haar-measure-preserving reflection $t = y - z$.

▼ Proof steps

1. **Apply the derivative theorem with its hypotheses.** The normalized bump φ_k is C^1 and compactly supported, while v is locally integrable on all of E . The compact-kernel convolution differentiation theorem therefore yields

$$Dv_k(y) = \int_E v(y - t) D\varphi_k(t) d\mu(t).$$

This is an integral of continuous linear functionals on E . The derivative $D\varphi_k$ is continuous and compactly supported. Multiplication against the locally integrable translate of v is therefore integrable, which justifies this operator-valued integral.

2. **Evaluate and change variables.** Evaluation $L \mapsto L(e)$ is continuous linear, so it commutes with that integrable Bochner integral:

$$Dv_k(y)e = \int_E v(y - t) D\varphi_k(t)e d\mu(t).$$

Translation and negation preserve additive Haar measure on the real vector space. Thus the measurable bijection $z \mapsto y - z$, its own inverse, permits substitution $t = y - z$. Since $y - (y - z) = z$, the preceding formula becomes

$$Dv_k(y)e = \int_E v(z) D\varphi_k(y - z)e d\mu(z).$$

No derivative of v has been used.

Main citations

- Exact declaration and its source — `MathlibAnnex.WeakGradient.localMollification_fderiv_apply`
- Trace of a rank-one endomorphism — `MathlibAnnex.trace_smulRight`
- Local integrability gives integrability on the compact carrier — `MathlibAnnex.WeakGradient.integrableOn_localBallCarrier`
- Global integrability of the compact localization — `MathlibAnnex.WeakGradient.integrable_compactLocalization`
- The shrinking bump with prescribed radii — `MathlibAnnex.WeakGradient.shrinkingBump`
- Inner radius of the shrinking bump — `MathlibAnnex.WeakGradient.shrinkingBump_rIn`
- Outer radius of the shrinking bump — `MathlibAnnex.WeakGradient.shrinkingBump_rOut`
- Local mollification as convolution — `MathlibAnnex.WeakGradient.localMollification`
- Normalization of the shrinking bump — `MathlibAnnex.WeakGradient.normalizedShrinkingBump`

Lean source signature (exact)

```
theorem localMollification_fderiv_apply
  (R : ℝ) (hR : 0 < R) {v : E → ℝ}
  (hv : LocallyIntegrable v μ) (k : ℕ) (y : E) (e : E) :
  fderiv ℝ (localMollification (μ := μ) R hR v k) y (e) =
    ∫ z, v z *
      fderiv ℝ (normalizedShrinkingBump (μ := μ) R hR k) (y - z) (e) ∂μ
```

In the source	Mathematical meaning
$[NormedAddCommGroup\ E]\ [NormedSpace\ \mathbb{R}\ E]$ $[FiniteDimensional\ \mathbb{R}\ E]\ [MeasurableSpace\ E]\ [BorelSpace\ E]\ \{\mu : Measure\ E\}$ $[\mu.IsAddHaarMeasure]$	The finite-dimensional real normed space E , with Borel measurable structure and additive Haar measure μ . Dimension zero is allowed.
$(R : \mathbb{R})\ (hR : 0 < R)\ \{v : E \rightarrow \mathbb{R}\}\ (hv : LocallyIntegrable\ v\ \mu)$	The positive radius R and locally integrable scalar function v . No derivative of v is assumed.
$(k : \mathbb{N})\ (y : E)\ (e : E)$	The kernel index k , point y and direction e .
$normalizedShrinkingBump\ (\mu := \mu)\ R\ hR\ k$	The normalized kernel φ_k , with inner radius $R/(2(k+2))$, outer radius $R/(k+2)$ and integral one, in the cited definitions.
$localMollification\ (\mu := \mu)\ R\ hR\ v\ k$	The scalar mollification $v_k(y) = \int_E \varphi_k(t) v(y-t) d\mu(t)$.
In the source	Mathematical meaning
$fderiv\ \mathbb{R}\ (localMollification\ (\mu := \mu)\ R\ hR\ v\ k)\ y\ (e)$	The derivative $Dv_k(y)$ evaluated on e , a real number.
$\int z, v\ z * fderiv\ \mathbb{R}\ (normalizedShrinkingBump\ (\mu := \mu)\ R\ hR\ k)\ (y - z)\ (e)\ \partial\mu$	The output formula $Dv_k(y)e = \int_E v(z) D\varphi_k(y-z) e d\mu(z)$. The derivative falls on the smooth kernel, not on v .

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 11–17:

```
noncomputable section
open Set Metric MeasureTheory Filter TopologicalSpace ContinuousLinearMap
open scoped ENNReal Topology Convolution BigOperators
namespace MathlibAnnex.WeakGradient
variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [FiniteDimensional ℝ E] [MeasurableSpace E] [BorelSpace E]
variable {μ : Measure E} [μ.IsAddHaarMeasure]
```

▼ **Exact content identity**

Declaration: `MathlibAnnex.WeakGradient.localMollification_fderiv_apply`

Accepted content SHA-256: 53fd32f1ec326a98c2b1dcfeaff7b1fc8e8743f6b690592ea00cc15721e7e43f

Accepted source guide SHA-256: 575b274040370cb823dabc732ad4a92e8adf030582ff26e29bb3a7fcb6ebdc75

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73